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Design And Development of API X65 Grade ERW Linepipe for 100% Pure Gaseous Hydrogen Transportation

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Low carbon, micro-alloyed (MA) with Ti/Nb/V/Mo in judicious combinations, are used to make linepipe steels, which are reliable mode of transportation of natural gas and hydrogen, over large distances under moderate to high pressure [1-2]. These steel and microstructures derived, are known to have excellent combination of strength, low temperature impact toughness, running fracture under dynamic loading conditions, fracture toughness, resistance to hydrogen induced cracking under acidic environment with excellent weldability [1-4].

In Indian scenario, majority of the linepipes are made of API 5L X52M-X70M grade steel, which is currently being used for natural gas transportation under non-sour environment. To reduce the economic aspects of hydrogen transportation and avoid cost of replacement, many non-sour pipelines are being considered for 5-15% blended hydrogen transportation. However, it is to be borne in mind that these pipeline steels are subjected to material degradation due to high pressure of hydrogen over time, commonly known as Hydrogen Embrittlement (HE) under stress. Such considerations are important as hydrogen transportation gains much attention as a sustainable energy carrier due to its high energy density and clean emissions profile [5-7].

Hydrogen induced cracking is known to be a critical phenomenon, that occurs due to the diffusion of hydrogen atoms into the steel and subsequent interactions with the microstructural features, which may cause a significant deterioration in ductility, impact and fracture toughness [8-11].

Introduction

Low carbon, micro-alloyed (MA) with Ti/Nb/V/Mo in judicious combinations, are used to make linepipe steels, which are reliable mode of transportation of natural gas and hydrogen, over large distances under moderate to high pressure [1-2]. These steel and microstructures derived, are known to have excellent combination of strength, low temperature impact toughness, running fracture under dynamic loading conditions, fracture toughness, resistance to hydrogen induced cracking under acidic environment with excellent weldability [1-4].

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Owing to the high diffusion coefficient of hydrogen atom in ferritic steel, hydrogen atoms may accumulate at various defect sites within a crystal, known as "trapping sites", such as microstructural defects, like point defects like vacancies, line defects like dislocations and other 3-dimensional defects like grain boundaries and interfaces of non-metallic inclusions with the matrix[12-13]. These trapping sites are either, "reversible" or "irreversible" types, depending on the negative binding energy of the trap with hydrogen atom [14]. Reversible trapping typically occurs sites having low-energy, like dislocations and grain boundaries, in which hydrogen atoms are weakly bound and therefore, are released without the necessity of any external energy. Grain boundaries and coincidence site lattice(CSL) boundaries, can either increase or decrease the diffusion rate of hydrogen atoms, by acting as reversible hydrogen trapping sites at the nodes [15-20]. Irreversible trapping happens at relatively higher negative energy sites, such as interfaces between precipitates and matrix, where hydrogen atoms are deeply bound and are not easily released without supply of external energy, like heat or strain. Amongst all irreversible trapping sites, carbides and their interfaces with matrix, are known for the strong trapping efficiency [21].

Variations in microstructural, crystallographic texture, precipitate morphology and therefore, mechanical properties across the thickness are common in thicker hot rolled plates, owing to gradient in deformation strain, cooling rates during hot rolling and strain increment during pipe forming [22-24].

Surface of hot rolled steels often contains large fraction of high angle grain boundaries [HAGB] compared to mid-thickness of the coil, which has a profound effect on the variation in tensile and fracture resistance under hydrogen environment [25], where surface typically exhibit a higher strength with relatively lower ductility compared to center showing moderate strength with improved ductility [22]. However, contractor views exists, where strength and hardness were found to initially decrease from the surface, attain a minima near quarterline before increasing again to the centerline of the thickness [26].

The variation in microstructure, texture and precipitation, across the thickness has a marked effect on the damage caused by hydrogen embrittlement. Coherent and semi-coherent interfaces with inclusions and carbides or carbonitrides with the matrix affect the degradation of mechanical properties in *bcc-Fe*, due to various degree of hydrogen trapping at their respective sites [27-29]. Regions with higher dislocation density, closer to the welded, surface-treated or heat affected zones, act as preferential sites for hydrogen trapping, predominantly in a reversible manner and therefore are susceptible to crack initiation and propagation [30-32]. Service conditions of pipelines over extended period of time, further results in operational stresses and environmental exposure, which leads to the aging related phenomena in hydrogen induced cracking [33-34].

In 2020, India's hydrogen demand stood at 6 MTPA. It is estimated that by 2030, the hydrogen costs will be down by 50%. The demand for hydrogen is expected to see a five-fold jump to 28 MT by 2050 where 80 per cent of the demand is expected to be green in nature. Under the scope of National Hydrogen Mission, Government of India aims to transform India into an energy independent nation by 2047 where green hydrogen will play an active role as an alternate fuel to petroleum/ fossil-based products. In 2022,

Tata Steel & Welspun Corp Limited, to a joint program to start development of API X65 grade steel for pipelines of pure H₂ & blended H₂ transportation at 100 bar pressure. This paper aims to address the design and developmental journey of API X65 grade steel and 14" ERW pipe made from the same, for 100% pure gaseous hydrogen transportation at 100 bar pressure, complying to the all critical requirements fulfilling ASME B 31.12. The observed results were explained in the light of the microstructures, transformation textures, measured from the orientation distribution function (ODF) along with size, distribution and coherency of extremely fine microalloyed precipitates with surrounding ferritic matrix and the subsequent results on interaction with hydrogen atoms.

Materials

Thermodynamic Calculations

Thermo-Calc software (using the thermodynamic database TCFE7) was used to calculate the equilibrium phase fractions and phase compositions of the given alloy as a function of temperature and are shown in Fig.1. The weight fraction of niobium, titanium and vanadium dissolved in austenite was determined from these calculations and is shown. The first phase to precipitate from the liquid steel[LIQUID] is δ -ferrite[BCC-A2] along with Ti-rich nitride precipitate [FCC-A1#2], which starts forming at 1500°C and remains compositionally stable up to 850°C, below which Ti/Nb-rich carbide precipitates starts to nucleate over TiN. It is interesting to note that, where Ti continues to precipitate out as Ti(C,N) in austenite till almost exhausted at 1050°C, Nb remains in solution of austenite[FCC-A1#1] till 1050°C, below which it starts precipitating out as complex carbonitrides. As austenite[FCC-A1#1] starts forming at 1425-1450°C, at the consumption of δ -ferrite[BCC-A2], the next two precipitates to form in austenite[FCC-A1#1] are, MnS[MS_B1] and Nb(C,N)[FCC_A1#3], which forms at 1280°C and 1080°C, respectively. As BCC-ferrite[BCC-A#2] evolves at 866°C at the consumption of austenite[FCC_A1#1], aluminium nitride[AlN_B4] starts precipitating, followed by complex V(C,N) precipitate[FCC_A1#4], followed by cementite[CEMENTITE_D011]. Other complex precipitates to form in bcc-ferrite[BCC-A2] are Cr₇C₃[M7C₃_D101] and Cr₂₃C₆[M23C6_D084], respectively.

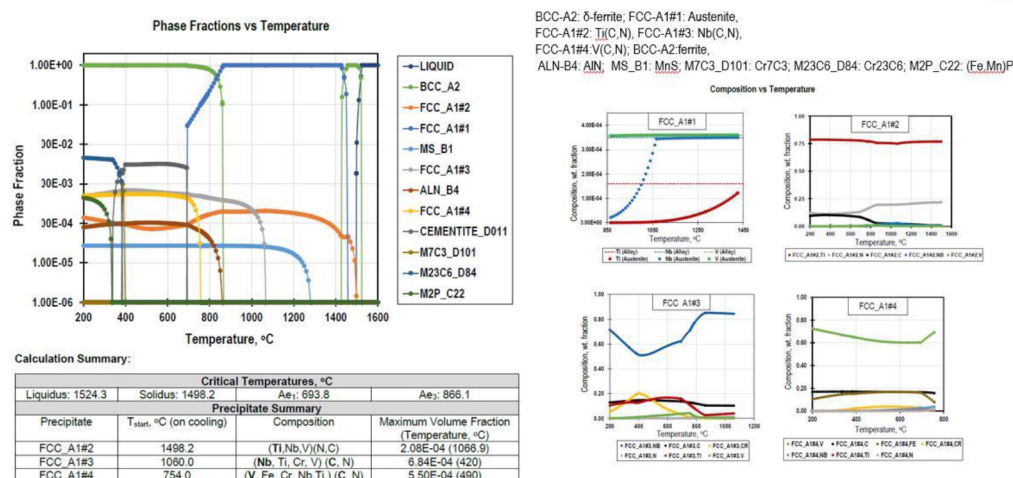


Figure 1—Thermodynamic calculation for the evolution of various carbides/ carbo-nitride precipitates and phases as a function of temperature under equilibrium. FCC-A1#-2, FCC-A1#-3 and FCC-A1#-4 are Ti-rich, Nb-rich and V-rich complex carbonitride precipitates, respectively.

Steel

The steel under investigation is a low-Carbon Cr+Ni+Nb+V based API X65 grade, suitable for 100% pure gaseous hydrogen transportation at high pressure [100 bar] though a 14" diameter ERW pipe. The steel was

made through deep desulphurisation, basic oxygen steelmaking followed by deep vacuum degassing [DS-BOF-OLP-LF-RH-Caster route] and subsequently thermo-mechanically rolled [TMCP] to make API X65 grade hot rolled steel. 14" ERW pipes were then made from the hot rolled coil produced and were subjected to test for HIC, SSC, SOHIC, CTOD at -29° along with K1H test as per ASTM E1681 under 100% pure H₂ at 100 bar pressure.

Hot Rolling

Continuously cast steel slab from the conventional twin-stand caster were cold charged in the reheating furnace where the temperature was maintained at 1220 - 1250 °C for homogenization of composition and microstructure, at the same time keeping the microalloying elements such as Ti, Nb and V, apart from Mn, Cr and Ni, in the austenite solid solution. Following descaling in high-pressure water jets, the slabs were then passed through the series of reverse rough rolling with decreasing rolling reductions in each subsequent stand of the roughing mill. The stand-wise average mill load and the mean flow stress (MFS) of austenite, as calculated using Sim's approach. There is a continuous increase in the MFS value observed from Strand-1 upto Strand-7 with values exceeding 200MPa beyond Strand-3. The hot rolling was conducted with target finish rolling temperature, T_{FRT} , aiming between $Ae3 - 50^{\circ}C < T_{FRT} < Ae3 + 50^{\circ}C$, where, $Ae3$ is the equilibrium austenite to ferrite transformation. A faster near group cooling strategy was adopted to reach at an intermediate temperature, T_{INT} , $Ae3 - 200^{\circ}C < T_{FRT} < Ae3 + 250^{\circ}C$, followed by correction header openings to finally achieve the target coiling temperature of 600°C.

Experimental Procedure

- a. Optical Metallography and SEM: The hot-rolled plate was sectioned using a SiC disc-cutter. The RD-ND orientation was mounted in conductive bakelite, polished and etched with 2% nital at 23°C for subsequent metallographic studies. Optical microscopy for quantitative metallography was carried out using Leica Zeiss upright light optical microscope in bright field mode. The average ferrite grain size and aspect ratio was measured through equivalent circle diameter (ECD) method encompassing more than 500 grains from each sample. The secondary electron images were then taken in Zeiss Supra25 FEG-SEM, operated at 15keV potential with working distance of 11 mm.
- b. Electron Backscatter Diffraction: Electron back scattered diffraction analysis, EBSD, was carried out using HKL channel 5 system attached to Zeiss Supra 25 FEG-SEM at a step size of 0.5 µm. During acquisition, the rolling direction of the sample was kept aligned with physical X-axis of the SEM and the scan was taken from the mid-section of the RD-TD plane, i.e. at half the thickness of the hot rolled plate and approximately 350 µm × 250 µm area was scanned. The acquired data was analysed with the help of TANGO, MAMBO and SALSA (Software provided with HKL system) to generate Band Contrast (BC), Grain Size Distribution, Inverse Pole Figures (IPF) and Orientation Distribution Functions (ODF) and Serial Sections respectively. $\Phi 2=45^{\circ}$ section of the serial sections was considered from the measurement of crystallographic texture fibers and relative intensities of the individual crystallographic orientations.
- c. Transmission Electron Microscopy: Transmission electron microscope (TEM) samples were prepared from the hot rolled specimen using wire-electrical discharge machining (EDM). The samples were then mechanically polished to a thickness of ~ 70 µm, followed by punching out $\Phi = 3$ mm discs. Subsequent thinning of the samples were carried out using twin jet polishing, in a Struers electropolisher with a fresh 10:90 mixture of perchloric acid and ethanol as the electrolyte under a voltage of ~ 21V at a subzero temperature of -29°C. This procedure ensured the thinning of the sample in the center without corroding, providing the necessary conditions for TEM observation. TEM analysis was conducted on the thinned samples using a JEM 3200FS operating at 300 kV.

Results and Discussions

Mechanical properties

The steel developed exhibits > 250 J of impact toughness upto -40°C with 100% shear area [% SA] along the RD, 45° to TD and TD to rolling, as shown in Fig.2. Drop weight tear tests [DWTT] was conducted as per API RP 5L3 drawing samples in (a) transverse to the rolling direction and (b) 45° to the rolling direction. One set [= 2 samples] of samples, each, were tested at 0°C , -20°C , -30°C and -40°C , respectively. The results of the %SA are reported in Table:1. The same has also exhibited $k_{IH} > 85$ MPa $\sqrt{\text{m}}$ under 100 bar pressure of gaseous hydrogen exposed for 1000 hours.

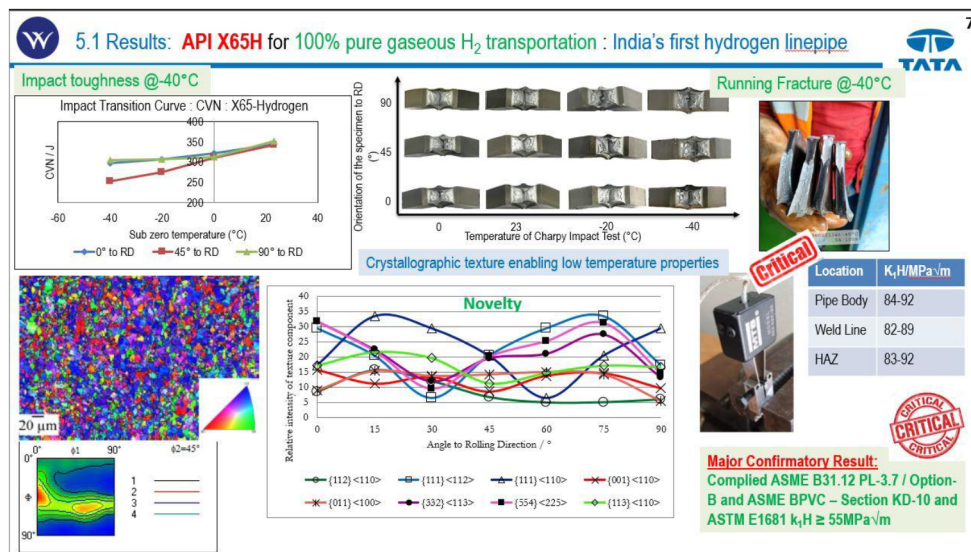


Figure 2—Charpy impact transition curves showing > 250 J of impact toughness up to -40°C in RD, 45° and TD to rolling direction. A running fracture %SA $> 95\%$ at -40°C accompanied by > 80 MPa $\sqrt{\text{m}}$ of k_{IH} test value as per ASTM E1681. These are attributed to various factors including microstructure, crystallographic texture and precipitate morphology and degree of coherency.

Table 1—DWTT % SA against testing temperature for API X65-Hydrogen grade developed

Testing temperature, $^{\circ}\text{C}$	%SA
0°C	100%
-20°C	100%
-30°C	100%
-40°C	100%

Optical Metallography

Fig.3 shows the optical image of the developed microstructure at $500\times$, revealing predominantly polygonal and quasi-polygonal ferrite with less than 7% 2nd phase as cementite. It is interesting to note a zero severity of pearlite banding as centerline segregation in the 50% depth sample, indicating effective dynamic soft reduction along with a stable and uniform casting speed and superheat control.

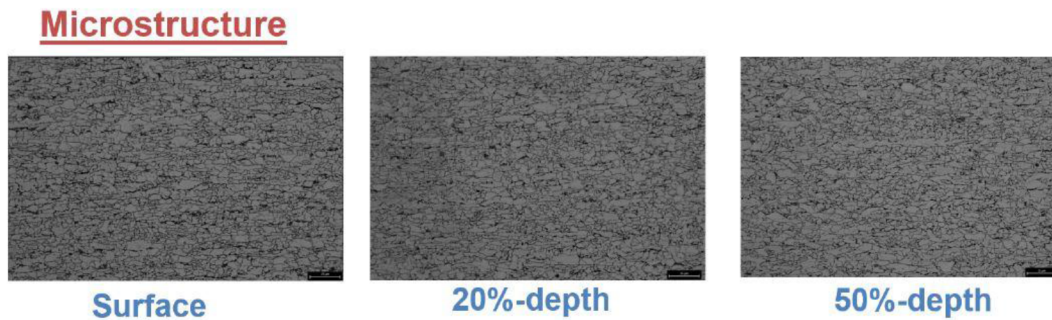


Figure 3—Optical images of the microstructure taken at surface, 20% depth and 50% depth from surface, showing predominantly polygonal and acicular ferrite with < 5% 2nd phase.

Secondary Electron Images

The secondary electron images, taken in Zeiss Supra25 FEG-SEM, operated at 15keV potential with working distance of 11 mm, are shown in Fig.4. The microstructure shows polygonal and quasi-polygonal ferrite with clear grain boundaries with insignificant amount of 2nd phase visible at 2500× and 5000×.

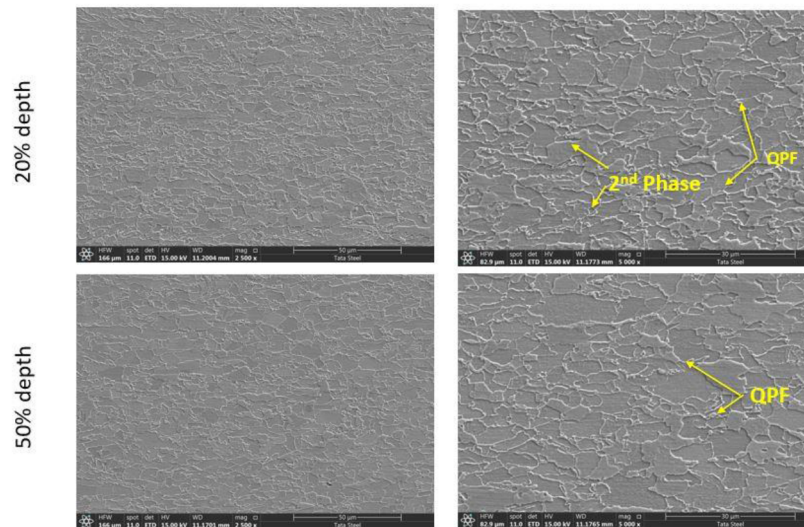


Figure 4—Secondary electron imaging of X65Hydrogen in RD-ND plane, at 20% depth and 50% depth

EBSD Results

The effect of $\{110\}$ family of planes are considered to be highly effective in increasing impact toughness in ductile mode of fracture, as the $\{110\}\langle 111 \rangle$ is the major slip system for plastic deformation in *bcc*-Fe lattice. Any components of crystallographic texture, that favors the activation of $\{110\}\langle 111 \rangle$ slip system, are believed to increase the impact toughness in the ductile regime. The contributions from $\{111\}\langle 110 \rangle$ and $\{111\}\langle 112 \rangle$ are very strong, more than 3x Random, with little difference in the intensities observed, contributed significantly in increasing the impact toughness, particularly at sub-zero temperature regime. There is a strong counter-active balance in fraction of grains with $\{111\}\langle 110 \rangle$ and $\{111\}\langle 112 \rangle$ texture observed at 30° and 60° to RD, further reducing anisotropy and upholding the ductile fracture behavior at lower temperature at all possible directions. This could possibly explain, the observed increase in impact toughness value at 45° to RD for at all the temperatures. Interesting observations were made with the $\{332\}\langle 113 \rangle$ and $\{554\}\langle 225 \rangle$ components, more than 3x random, which are responsible for reducing the toughness anisotropy in ductile/ductile-brittle mode of fracture. The hot band texture showed an increased trend of $\{332\}\langle 113 \rangle$ and $\{554\}\langle 225 \rangle$ component. Large fractions of grains with $\{332\}\langle 113 \rangle$ and $\{554\}\langle 225 \rangle$ component over $\{113\}\langle 110 \rangle$ and $112\langle 110 \rangle$ kept the extent of anisotropy low any directions

over all the temperatures tested. High impact toughness of the alloy is attributed to a strong $\{111\}\langle 110\rangle$ and $\{111\}\langle 112\rangle$ texture across all the directions. Moreover, the volume fraction of $\{001\}$ grains parallel to the main fracture plane is very less leading to increased cohesiveness.

Crystallographic texture: Thermo-mechanical controlled rolling below T_{nr} [n-recrystallisation] temperature of austenite leads to the development of α -fiber with strong $\{113\}\langle 110\rangle$ and $\{112\}\langle 110\rangle$ components. Sharpening of these components results in increased impact toughness values, although a stronger presence of $\{112\}\langle 110\rangle$ over $\{332\}\langle 113\rangle$ could have resulted in significant impact anisotropy with respect to the rolling direction. A stronger γ -fiber, more than 3x random, formed a result of (a) higher Nb content in the steel along with (b) larger reduction in $(\gamma+\alpha)$ region at later stands of the finishing mill, which has possibly increased both the $\{111\}\langle 112\rangle$ and $\{111\}\langle 110\rangle$ components. This is further confirmed by strong presence of $\{223\}\langle 110\rangle$ at $\phi=43^\circ$ of the α -fiber.

Toughness Anisotropy: Total contribution of the γ -fiber components, namely, $\{111\}\langle 112\rangle$ and $\{111\}\langle 110\rangle$ over the $\{112\}\langle 110\rangle$ and $\{113\}\langle 110\rangle$ of the α -fiber, plays a significant role in increasing both the impact toughness value and mode of fracture to ductile regime. The major factor reducing the anisotropy in impact toughness is the higher fraction of grains with $\{332\}\langle 113\rangle$ and $\{554\}\langle 225\rangle$ component over $\{112\}\langle 110\rangle$ and $\{113\}\langle 110\rangle$, as shown in Fig.5. The steel also exhibits strong through-thickness texture components, where the centerline also exhibits more than 3x random strength of γ -fiber, as shown in Fig.6.

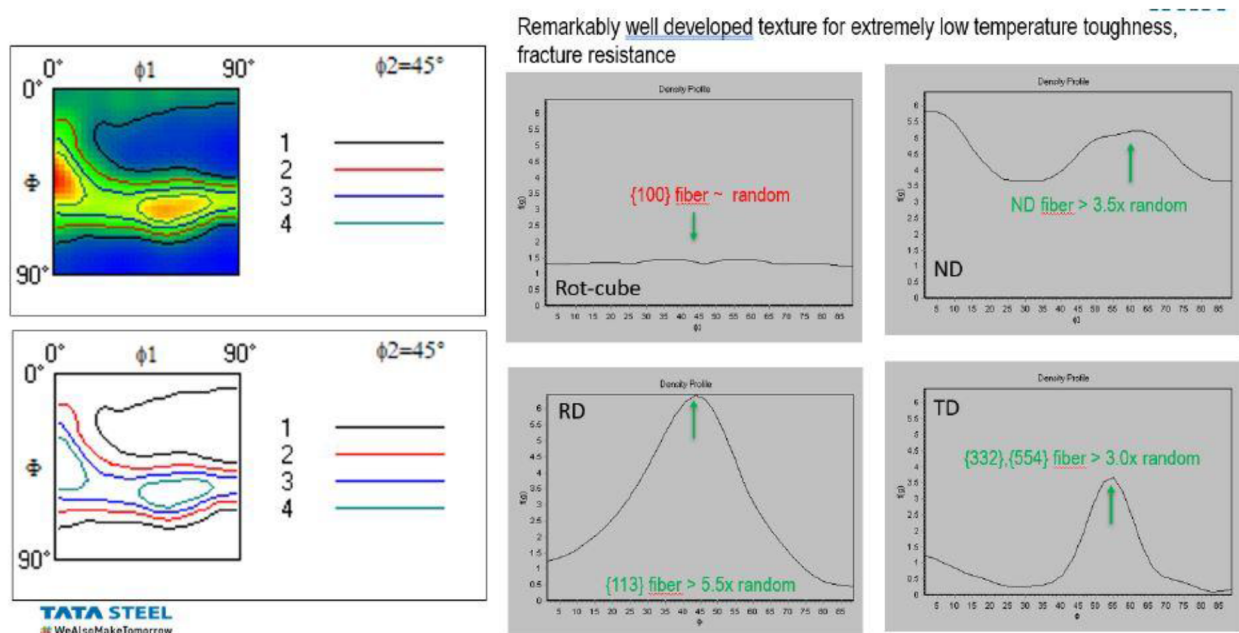


Figure 5—Fiber plots of X65Hydrogen in RD-TD plane, at 20% depth showing RD, ND and TD fibers.

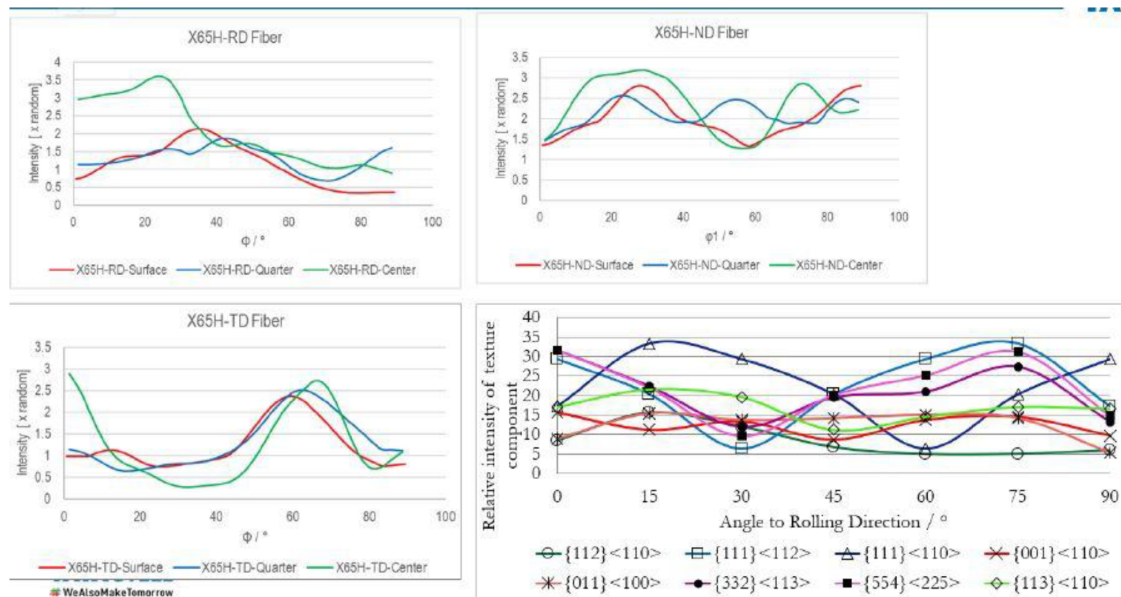


Figure 6—Skeleton plots of X65Hydrogen in RD-TD plane, at surface, 20% depth and 50% depth showing RD, ND and TD fibers.

TEM Results

The bright field TEM images [Fig.7] show polygonal ferrite, acicular/ bainitic ferrite and some degenerated pearlite in the microstructure. There are both fine and coarse precipitates of TiCN, NbCN, along with much finer VCN, which forms at lower temperature in Run-out-table of HSM. The SADP of precipitates along the zone axis of $B = [110]$ and $[012]$, diffracting in-plane with the matrix of $bcc\text{-}Fe$ shows fineness and in-plane formation of carbide precipitates.

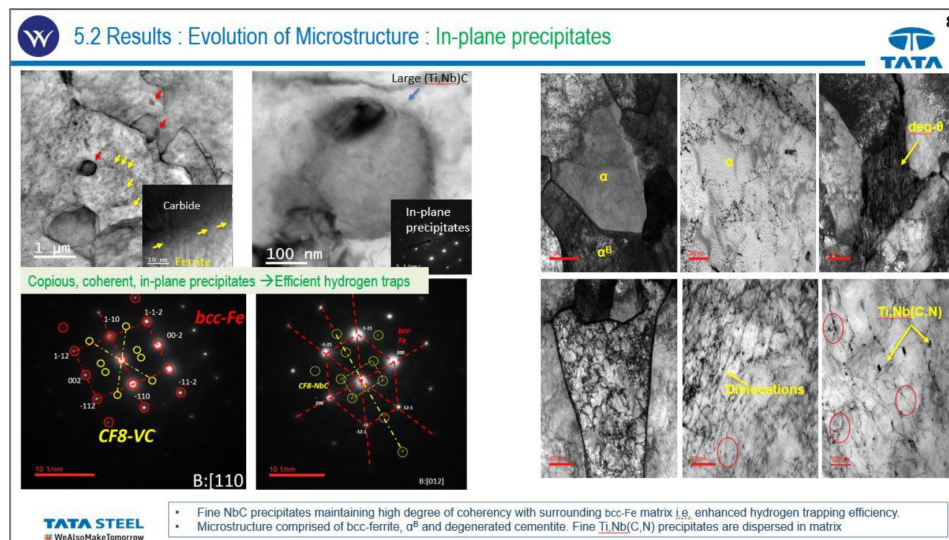


Figure 7—Bright field TEM image with SADP pattern showing in-plane precipitates of carbides with ferrite matrix with SADP of $bcc\text{-}ferrite$ and precipitate of NbC appearing under same zone axis of $B [110]$ and $B [012]$.

A fine VC precipitate of ~ 15 nm diameter with semi-coherent interface with surrounding $bcc\text{-}Fe$ matrix is shown in Fig. 8. It is believed that either the C-vacancy within the VC lattice or C-vacancy at the interface of precipitate / matrix acts as potential irreversible trapping site for atomic hydrogen. When values of $k1H$ are plotted against other conventional linepipe steels, the alloy under discussion exhibits significantly higher $k1H$ value of >83 $MPa\sqrt{m}$, substantiating the effect of microstructural features.

- c. Solution: 5wt% NaCl + 0.5 wt% glacial acetic acid in distilled water.
- d. pH range of the solution: 2.70+/-}0.1.
- e. rate of N₂ purging: 100 ml/min/ lit.
- f. rate of H₂S purging: 200 ml/min/ lit.
- g. duration of H₂S purging: 1 hour.
- h. duration of test: 96 hours.
- i. pH after H₂S concentration: 3.3 max.
- j. H₂S concentration: 2300 - 3500 ppm.

The test for evaluation of resistance to hydrogen-induced cracking need to meet the following acceptance criteria, with each ratio being the maximum permissible average for three sections per test specimen when tested in Solution (Environment) A (ref ISO 15156-2, Table B.3):

- a. Crack Sensitivity Ratio (CSR) $\leq 1.0\%$
- b. Crack Length Ratio (CLR) $\leq 10\%$
- c. Crack Thickness Ratio (CTR) $\leq 3\%$.

Values of crack-length ratio, crack-thickness ratio and crack-sensitivity ratios were reported and found to be zero.

Sulphide Stress Cracking [SSC]

The Sulphide Stress Cracking [SSC] test was conducted as per NACE TM0177 Method-C [four point bending] method. The salient feature of the test were;

Frequency: The SSC test i.e. (One set of three transverse specimens shall be taken from each coil shall be carried out at One test for each of the first three heats applied. [one test per heat, 2 heats only].

Test pieces for four-point bending: Test pieces for four-point bending SSC tests shall be ≥ 115 mm long $\times$ 15 mm wide $\times$ 5 mm thick. Samples shall be prepared from the surface which is supposed to become the inner surface of the pipe and it shall be in tension during test. Noted. Sample will be taken from tail end (OD) at 1/4th position of width. Total 3 specimens were taken per heat from tail end (OD), each at the 1/4 position of width, at 1/2th position of width and at 3/4 th position of width).

- SSC tests shall be performed as follows:
 - the test method solution control shall be in accordance with NACE TM0177 Method C.
 - the test piece shall be as defined in ASTM G39.
 - the test solution shall be Solution A as defined in NACE TM0177.
 - H₂S Concentration: 2300-3500 ppm.
 - Final pH of the solution: 4.0 max
 - the test duration shall be 720 hours.
 - the test pieces shall be bent using Four Point Loading Jigs to be stressed to minimum 0.90

times AYS and then immersed in Solution A as defined in NACE TM0177.

After removal of the SSC test specimens from the test medium, the specimen surface previously under tension were examined under a low-power microscope at X10 magnification. The occurrence of any surface breaking fissures or SSC cracks on the tension surface of the test specimen constitute failure of the specimen unless it can be demonstrated that these are not the result of sulphide stress cracking. There were no SSC cracks or blisters observed in any of the specimens tested.

Stress Oriented Hydrogen Induced Cracking [SOHIC]

The material's resistance to Stress Oriented Hydrogen Induced Cracking [SOHIC] was conducted as per NACE TM 0177. No ladder-like HIC or cracks of any form were observed.

Crack Tip Opening Displacement

The crack tip opening displacement [CTOD] test on pipe body was conducted as per BS 7448 part-1 standard and ISO 12135. Test pieces were taken and prepared in accordance with ISO 12135. Sample were drawn from ¼ th position from the width of the coil. Total three specimens were taken per heat from tail end (OD), each at ¼ th position of width, at 1/4th position of width and at ¾ th position of width. The location of the fatigue crack tip were verified by post-test sectioning and metallography. Only specimens that are qualified with respect to crack tip location by post-test metallographic examination were considered valid. One set of test pieces [= 3 specimens], with each set consisting of three test pieces, were extracted from the coils for Single Edged Notched Bend (SENB) specimen in transverse direction.

Testings were conducted on through thickness notched specimens in accordance with ISO 12135 at -29°C. Minimum acceptable critical CTOD value shall be 0.25 mm for specimens. Following information from CTOD testing were reported; (a) Load vs. crack mouth opening displacement curves of all tests, (b) Crack measurements, (c) δ -results. The results as shown in Fig.10 shows average CTOD value of 1.50 mm at -29°C [against a target of 0.40 mm average at -29°C] over three specimens drawn from pipe body.

TEST CERTIFICATE

To
TATA STEEL LIMITED
R&D Building, Jamshedpur 831007.

T.C. NO : **2209409**
Date : **08/12/2022**
Ref Date : **28/10/2022**
Date of Receipt : **01/11/2022**

CTOD TEST AS PER : BS 7448 Part 1
WORK ORDER DETAILS:-3000116298/132.

Description : CTOD TESTING OF SAMPLES SIZE 400 X 400 mm

Geometry	BEND		
Crack Plane Orientation	PARENT METAL		
Sample ID	K279930903-1	K279930903-2	K279930903-3
Thickness (B) [mm]	10.01	10.03	10.02
Width (W) [mm]	20.04	20.02	20.06
Notch Length (a ₀) [mm]	7.00	7.00	7.00
Span (S) [mm]	80.00	80.00	80.00
Yield Stress (σ _{ys}) [Mpa]	569.00	569.00	569.00
Ultimate Tensile Stress (UTS) [MPa]	619.33	619.33	619.33
Poissons Ratio	0.33	0.33	0.33
Modulus of Elasticity (E) [Gpa]	210.00	210.00	210.00
Terminal Precracking Force (P) [kN]	3.26	3.31	3.28
Load Ratio @	0.10	0.10	0.10
Precracking Temperature [°C]	RT	RT	RT
No of Cycles	55217.00	58425.00	53965.00
Final Crack Length (Δa+a) [mm] *	10.35	10.55	10.40
Test Temp [°C]	-29.00	-29.00	-29.00
Plastic part of COD opening (V _p) [mm]	3.18	3.79	3.52
Load corresponding to V _p , [kN]	10.88	10.93	10.52
Load line Displacement Rate [MPa√m/s]	1.25	1.32	1.27
CTOD Value (δ) [mm]	1.37	1.63	1.51
Average CTOD [mm]**	1.50		

Figure 10—The crack tip opening displacement test results of API X65Hydrogen at -29°C as per BS 7448 part-1.

ASTM E1681-K1H test: Critical Threshold Stress Intensity Factor

The critical threshold stress intensity factor was measures as per ASTM E1681 [Fig.11] and the results are listed below;

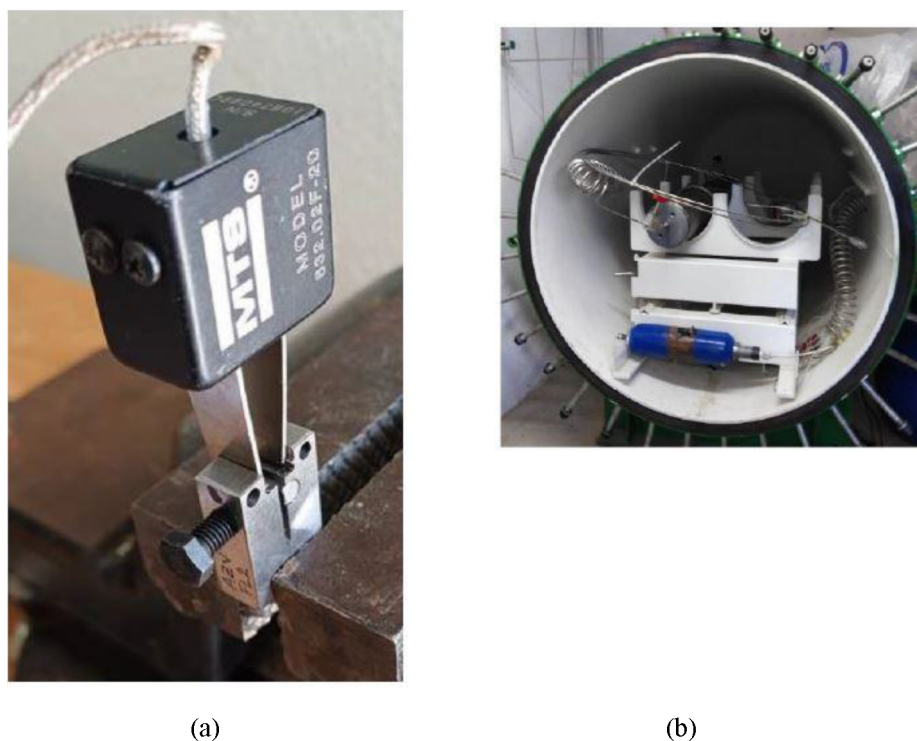


Figure 11—Photographs taken before ASTM E1681 K1H test, (a) bolt-loading of the specimen with clip-gauge attached and (b) pre-loaded samples put inside a sealed cannister kept at 100 bar pressure of 100% pure gaseous hydrogen.

For base material, the minimum value of KIH calculated was:

- a. $> 83.9 \text{ MPa}\sqrt{m}$ for the base metal and
- b. $> 87.3 \text{ MPa}\sqrt{m}$ for the weld centerline.

No significant microstructural difference between both materials and samples taken before and after 1000 hours of H₂ exposure were observed.

Hydrogen Induced Cracking [HIC] in Electric Resistance Weld [ERW] of linepipe steels: A brief review

High-frequency electric resistance welding, commonly known as HFW or ERW, is a popular pipe manufacturing process for lower wall thickness [$t = 5 - 19 \text{ mm}$] and pipe diameter [6-24 inches], for manufacturing of oil and gas transmission pipelines. The HRC is cold-formed and then electrically seam welded without any need of filler metal wire or fluxes. Therefore, it is essential that the weld joints produced in the ERW process should be free from oxide inclusions, which has got a deleterious effect on weld quality, namely strength and toughness for non-sour applications. When it comes to the sour-service application, the quality of weld, in terms of cleanliness and integrity, becomes a parameter of immense importance related to Hydrogen Induced Cracking [HIC] and Sulphide Stress Cracking [SSC]. One of the major concerns with ERW pipes is the way nonmetallic, heterogeneous inclusions and non-homogeneous microstructures formed during ERW, affect the HIC resistance along the bondline of the weld. The oxide formed during high-temperature of the ERW process, should be flushed out of the weldment by proper squeezing of the liquid weld metal pool during the welding process, while the oxide is still in the liquid-phase, as this ensures that the viscosity and surface tension of the liquid metal are kept to the minimum. Presence of any crystalline material in the weld metal pool sharply increases the viscosity of the melt resulting in improper squeezing. As a result, two parameters of the ERW process becomes crucial for good weld quality, (a) the temperature

of the ERW process should be high enough to lower the viscosity of the weld metal pool and (b) composition of the oxide should be such that it remains in the liquid state at the ERW temperature.

Centerline slitting of hot rolled coils are likely to cause poor HIC resistance of the ERW weld due to higher severity of centerline segregation [CLS] and increased level of inclusions in the weld and therefore, are strictly avoided. Considerable research has been made on the oxide-metal reactions during different welding processes and modelling of oxide formation. Formation of SiO_2 takes place before the generation of MnO in weld pool, however, the potential formation of complex oxides such as rhodonite or fayalite was not considered, as the temperature involved is too high for these phases to be stable during the welding. The formation of high temperature oxides is controlled by the $(\text{Mn}+\text{Cr})/\text{Si}$ ratio, which should be kept in the region $(\text{Mn}+\text{Cr})/\text{Si} < 6$, with an additional restriction that $4 \leq \text{Mn}/\text{Si} \leq 7$. Majority of high strength linepipe steels, namely API X65 and X70, are Fe-Mn-Si-Cr based alloys with addition of micro-alloying elements like, Ti, Nb and/or V [with or without addition of Mo and/or Ni, for low temperature toughness], it is imperative to study the oxide formation behaviour of the Fe-Mn-Si-Cr-O system at the prevailing temperature of the ERW process.

Conclusions. The developed API X65 grade steel suitable for 100% gaseous hydrogen transportation at 100 bar pressure, claimed to;

- a. be made within a thickness range of 5.0 - 14.0 mm, width range of 900-2000 mm in hot rolled coil stage, made from the hot strip mill of Tata Steel Kalinganagar. The steel was made through DS-BOF-OLP-LF-RH-Continuous Caster route and hot rolled coils were made through a 2100 mm wide 6.5MnT capacity hot strip mill, exhibiting a yield stress ranging ~ 525 MPa, ultimate tensile stress ranging between 605 MPa and Total Elongation >30% and yield ratio <0.90.
- b. the microstructure developed, is suitable to exhibit impact toughness [through Charpy V-notch] of >150J, between a temperature range of +23°C to -40°C, in hot rolled coil stage in rolling direction, 45° to rolling direction and transverse to rolling direction with >90% shear area. The running fracture in drop weight test fracture exhibits a % shear area of >95% between a temperature range of +23°C to -40°C at hot rolled coil stage in rolling direction, 45° to rolling direction and transverse to rolling direction.
- c. the microstructure exhibits a crack tip opening displacement [CTOD as per BS 7448 Part-1] value of >1.2 mm, between a temperature range of +23°C to -29°C, both at hot rolled coil stage and at the pipebody post pipe forming.
- d. the steel is resistance to hydrogen induced cracking [HIC] as per NACE TM0284:2016 in Solution "A" [pH range 2.7 to 3.3] with CLR <10%, CTR <3% and CSR <1% in TDC guarantee and CLR=0%, CTR=0% and CSR=0% in actual material in (a) hot rolled coil condition, (b) ERW pipe weld flow line, (b) ERW pipe weld flow line + 2 mm and (b) ERW pipe weld flow line + 4 mm. No blisters were observed.
- e. resistance to sulphide stress cracking [SSC] as per NACE TM0177:2016, Method -B, ISO 15156-2 and ASTM G39-99 in Solution "A" is also observed with 90%AYMS, with no indication of SSC cracks or Ladder-like HIC crack initiation under stress oriented hydrogen induced cracking [SOHIC] condition. This observation is made in (a) hot rolled coil condition, (b) ERW pipe weld flow line, (b) ERW pipe weld flow line + 2 mm and (b) ERW pipe weld flow line + 4 mm.
- f. when it comes to fracture toughness under high pressure hydrogen environment, the developed microstructure is capable of exhibiting >80 $\text{MPa}\sqrt{\text{m}}$ of K_{IH} value after exposure to 100% pure gaseous hydrogen for 1000 hours as per ASTM E1681 at (a) pipe body and (b) ERW weld flow line. The K_{IApp} exceeded 165 $\text{MPa}\sqrt{\text{m}}$ in all specimens.

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